

CSE370 : Database Systems

Assignment - 03 | Spring 2025

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No 1 Query (as Plain Text)	select customer.customer_name, loan.loan_number from borrower join loan on borrower.loan_number=loan.loan_number and loan.branch_name="Downtown" join customer on borrower.customer_id= customer.customer_id;
No 1 SS (of Query & Output in Shell)	<pre>MariaDB [Bank_22201403]> select customer.customer_name, loan.loan_number from -> borrower join loan on borrower.loan_number=loan.loan_number -> and loan.branch_name="Downtown" join customer -> on borrower.customer_id= customer.customer_id; +-----+-----+ customer_name loan_number +-----+-----+ Johnson L-14 Jones L-17 Williams L-17 +-----+-----+ 3 rows in set (0.001 sec)</pre>

No 2 Query
(as Plain Text)

```
select c1.customer_name as customer1,  
c2.customer_name as customer2, c1.customer_city  
as city from customer c1 join customer c2  
on c1.customer_city=c2.customer_city  
and c1.customer_id<c2.customer_id;
```

No 2 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_22201403]> select c1.customer_name as customer1,  
-> c2.customer_name as customer2, c1.customer_city  
-> as city from customer c1 join customer c2  
-> on c1.customer_city=c2.customer_city  
-> and c1.customer_id<c2.customer_id;
```

```
+-----+-----+-----+  
| customer1 | customer2 | city      |  
+-----+-----+-----+  
| Jones     | Hayes     | Harrison  |  
| Smith     | Curry     | Rye       |  
| Lindsay   | Adams     | Pittsfield|  
| Turner    | Green     | Stamford  |  
+-----+-----+-----+  
4 rows in set (0.001 sec)
```

No 3 Query
(as Plain Text)

```
Select Account.branch_name as Branch_name,  
SUM(Account.balance*(4/100)) as  
Total_interest from account  
Group by Account.branch_name;
```

No 3 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_22201403]> Select Account.branch_name as Branch_name,  
-> SUM(Account.balance*(4/100)) as  
-> Total_interest from account  
-> Group by Account.branch_name;
```

```
+-----+-----+  
| Branch_name | Total_interest |  
+-----+-----+  
| Brighton    | 66.0000        |  
| Downtown    | 20.0000        |  
| Mianus       | 28.0000        |  
| Perryridge  | 16.0000        |  
| Redwood     | 28.0000        |  
| Round Hill  | 14.0000        |  
+-----+-----+  
6 rows in set (0.001 sec)
```

No 4 Query
(as Plain Text)

```
Select b.branch_city, a.account_number, a.balance
From account a Inner Join branch b on a.branch_name = b.branch_name
Where a.balance = ( Select MAX(a2.balance) From account a2
Inner Join branch b2 on a2.branch_name=
b2.branch_name Where b2.branch_city=
b.branch_city) Order By b.branch_city;
```

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_22201403]> Select b.branch_city, a.account_number, a.balance
-> From account a Inner Join branch b on a.branch_name = b.branch_name
-> Where a.balance = ( Select MAX(a2.balance) From account a2
-> Inner Join branch b2 on a2.branch_name=
-> b2.branch_name Where b2.branch_city=
-> b.branch_city) Order By b.branch_city;
+-----+-----+-----+
| branch_city | account_number | balance |
+-----+-----+-----+
| Brooklyn   | A-201          | 900     |
| Horseneck  | A-215          | 700     |
| Palo Alto  | A-222          | 700     |
+-----+-----+-----+
3 rows in set (0.006 sec)
```

No 5 Query
(as Plain Text)

```
Select * from ( Select loan.loan_number, amount, customer_name
from loan Inner Join borrower on
loan.loan_number=borrower.loan_number
Inner Join customer on customer.customer_id = borrower.customer_id
order by amount DESC limit 5 )
As table1 Order By amount, loan_number DESC;
```

No 5 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_22201403]> Select * from ( Select loan.loan_number, amount, customer_name
-> from loan Inner Join borrower on
-> loan.loan_number=borrower.loan_number
-> Inner Join customer on customer.customer_id = borrower.customer_id
-> order by amount DESC limit 5 )
-> As table1 Order By amount, loan_number DESC;
```

```
+-----+-----+-----+
| loan_number | amount | customer_name |
+-----+-----+-----+
| L-17        | 1000   | Jones         |
| L-16        | 1300   | Adams         |
| L-15        | 1500   | Hayes         |
| L-14        | 1500   | Johnson       |
| L-23        | 2000   | Smith         |
+-----+-----+-----+
```

```
5 rows in set (0.001 sec)
```

No 6 Query (as Plain Text)	Select distinct c.customer_name from customer c Inner Join depositor d on c.customer_id=d.customer_id Inner Join account a on d.account_number=a.account_number Inner Join borrower b on c.customer_id=b.customer_id Inner Join Loan l on b.loan_number = l.loan_number and l.branch_name=a.branch_name Where a.branch_name= "Perryridge";
No 6 SS (of Query & Output in Shell)	<pre> MariaDB [Bank_22201403]> Select distinct c.customer_name from -> customer c Inner Join depositor d on c.customer_id=d.customer_id -> Inner Join account a on d.account_number=a.account_number -> Inner Join borrower b on c.customer_id=b.customer_id Inner Join -> Loan l on b.loan_number = l.loan_number and -> l.branch_name=a.branch_name Where a.branch_name= "Perryridge"; +-----+ customer_name +-----+ Hayes +-----+ 1 row in set (0.001 sec) </pre>

No 7 Query
(as Plain Text)

```
Select c.customer_name, Count(*) as number_of_loans,  
SUM(L.Amount) As Total_loan From Customer c Join  
borrower b on c.customer_id=b.customer_id Join Loan l on  
b.loan_number=l.loan_number Where c.customer_id  
In(select b2.customer_id From borrower b2  
Group By b2.customer_id Having Count(*)>=2)  
Group By c.customer_id Order By Total_loan DESC;
```

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_22201403]> Select c.customer_name, Count(*) as number_of_loans,  
-> SUM(L.Amount) As Total_loan From Customer c Join  
-> borrower b on c.customer_id=b.customer_id Join Loan l on b.loan_number=l.loan_number Where c.customer_id  
-> In(select b2.customer_id From borrower b2  
-> Group By b2.customer_id Having Count(*)>=2)  
-> Group By c.customer_id Order By Total_loan DESC;  
+-----+-----+-----+  
| customer_name | number_of_loans | Total_loan |  
+-----+-----+-----+  
| Smith        | 2               | 2900       |  
+-----+-----+-----+  
1 row in set (0.001 sec)
```